

Food-intake detection: study design (internal)

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1 Proposed study design

The literature review points to one gap more than any other: food-intake detectors from wearable chewing sensors are usually reported as a single architecture with a single number, and the design decisions behind them are rarely isolated. The ablation in Table 1 already shows that those decisions are not equally important, since the input representation accounts for most of the performance and the attention components cannot be separated from noise. The proposed study makes that the question rather than a side result: which design dimensions matter for deep food-intake detection from a chewing sensor, measured one at a time under leave-one-subject-out validation.

Table 1: Preliminary ablation: the model with one component removed or added. The first 20 annotated sessions, participant-level 4-fold cross-validation, same folds for every row. Fold-to-fold spread in F1 was about 0.048, so effects smaller than that are not distinguishable from noise. To be repeated on all 61 sessions.

Variant	F1	Change	Precision	Verdict
Full model	0.854	–	0.876	–
FFT magnitude input instead of time series	0.767	–0.087	0.696	Hurts
No overlapping training windows	0.829	–0.025	0.834	Possibly hurts
No attention (global average pooling)	0.847	–0.007	0.857	Within noise
No chew-count head	0.849	–0.006	0.857	Within noise
Add median smoothing (3 windows)	0.775	–0.079	0.765	Hurts
Add 8 s context each side, no marker	0.775	–0.080	0.683	Hurts

1.1 Protocol, fixed for every experiment

- **Leave-one-subject-out validation** on all 43 participants (43 folds, each holding out every session of one participant), following Saeb et al. (1). At about a minute per fold this is computationally feasible, and it is the closest approximation of the use case: a participant the model has never seen. Five-fold runs are used only to explore.
- **Paired comparison per participant.** Each variant is scored on the same held-out participants as the reference model, so the difference can be tested per participant (Wilcoxon signed-rank)

rather than read against the spread of fold means.

- **Three seeds per configuration**, reported as mean and range, so initialisation noise is not mistaken for an effect.
- **False positives per hour of non-eating** reported alongside F1, because that is the failure mode that matters in free-living use and the one negative data (below) is meant to fix.

1.2 Dimensions to explore

Table 2: Planned experiments. Each changes one dimension of the model in the model and holds everything else fixed.

Dimension	Levels	Question	Grounding
Published architectures	FCN, ResNet and Inception as compared by Ghosh and Sazonov (2), trained on the same folds	Where does this model stand against the closest prior work on AIM-2, under an identical protocol?	Ghosh and Sazonov (2), Wang et al. (3)
Input representation	Time series, FFT magnitude, wavelet scalogram	Does keeping temporal structure explain the gain, and does a time-frequency image recover it?	Ghosh et al. (4), Hannun et al. (5)
Sampling rate	128, 64, 32, 16 Hz	How much of the signal is needed? Chewing lies below about 2.2 Hz, so 32 Hz still leaves a Nyquist margin of seven times.	Po et al. (6)
Window length	2, 4, 8, 16 s	Resolution against context: shorter windows localise bites, longer windows contain more chew cycles.	Banos et al. (7)
Context	None; 8 s each side with a target-window marker channel	Can surrounding signal help without the model answering "is there eating anywhere nearby?"	(this work)
Chew-head weight	0, 0.1, 0.2, 0.5, 1.0	How strongly should counting chews shape the features, and does it help participants with weak optical signal?	Caruana (8)
Temporal model	Attention pooling; small Transformer encoder over the 64 convolutional time steps	Does self-attention between time steps add to a convolutional front end at this data size?	Ikram et al. (9), Zerveas et al. (10), Zeng et al. (11)

Dimension	Levels	Question	Grounding
Self-supervised pre-training	None; convolutional trunk pre-trained on unannotated AIM-2 recordings in ETS	Can the unannotated sensor data, most of what the database holds, stand in for labels the study does not have?	Diou et al. (12)
Negative data	Annotated sessions only; plus non-eating recordings from other ETS studies	Do additional negative participants reduce false positives without costing recall?	Ghosh et al. (4), Bell et al. (13)

The published-architecture row comes first: it is the comparison a reviewer will ask for, and it replaces comparisons with the lab's earlier model. The representation, sampling-rate and window rows are cheap and answer the questions raised at the 9/23 lab meeting directly. The Transformer row is the one most likely to be asked about by reviewers, and the one most at risk of overfitting on a dataset this size; with 24 hours of annotated data it is now worth running, and should be reported whether it helps or not. Negative data is expected to have the largest practical effect, since false positives are the model's main weakness, and is the natural joint experiment with the lab.

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